

SEAN THOMAS

SENIOR DATA ENGINEER

seanthomas78600@gmail.com 325-240-5719 Branchburg, NJ 08853 /in/seany-thom-b884833b0

SUMMARY

Senior Data Engineer with 10+ years of experience designing and operationalizing enterprise-scale **Azure Databricks Lakehouse** platforms within **Big Data** environments across healthcare, fintech, and consulting domains. Deep expertise in **Data Pipeline Development**, **Delta Lake**, **Apache Spark**, **Structured Streaming**, and HIPAA-compliant distributed systems. Proven ability to architect **CDC-enabled Medallion Architectures (Bronze/Silver/Gold)**, implement enterprise **Data Quality Frameworks**, optimize large-scale Spark workloads, and deliver governed, **ML-ready data ecosystems**. Strong Azure-first and AWS-integrated cloud engineering background supporting secure, high-performance analytics platforms and enterprise BI enablement.

TECHNICAL SKILLS

- **Data Engineering & Pipelines:** Airflow (DAG orchestration, scheduling, retries, alerting), Python ETL pipelines, batch and incremental processing, ELT frameworks, legacy pipeline modernization, data ingestion, transformation, and orchestration, production pipeline support, failure recovery, backfills, dependency management
- **Cloud Platforms:** Microsoft Azure, AWS, Azure Databricks, ADLS Gen2, Azure Data Factory, Synapse Analytics, AWS S3, AWS Glue, IAM
- **Data Engineering:** Apache Spark, PySpark, SQL, Scala, Delta Lake, CDC, Structured Streaming, Data Pipeline Development
- **Lakehouse Architecture:** Medallion Architecture, Dimensional Modeling, Data Vault, Star Schema
- **Streaming & Messaging:** Kafka, Azure Event Hub, Schema Registry
- **Governance & Security:** Unity Catalog, RBAC, HIPAA, Encryption, Tokenization, Data Lineage, Data Quality Frameworks
- **Performance Optimization:** Adaptive Query Execution (AQE), Broadcast Joins, Partition Pruning, Z-Ordering, Shuffle Optimization
- **DevOps & Automation:** Azure DevOps, Git, CI/CD Pipelines, Terraform
- **BI & Analytics Enablement:** Power BI, Tableau
- **Data Formats:** Parquet, ORC, Avro, JSON
- **Programming:** Python, SQL, Scala

PROFESSIONAL EXPERIENCE

CDW

Senior Azure Databricks Data Engineer

JAN 2021 – PRESENT

Led enterprise Lakehouse modernization initiatives transforming legacy ETL ecosystems into scalable, governed Azure Databricks platforms processing 8TB+ daily workloads.

Architecture & Platform Design

- Architected enterprise-scale Azure Databricks Lakehouse Architecture using Delta Lake on ADLS Gen2,

reducing processing latency by 35% and enabling near real-time analytics.

- Designed and implemented Bronze/Silver/Gold Medallion Architecture, standardizing ingestion, curation, and consumption layers across distributed workloads.
- Defined enterprise data modeling standards using Dimensional Modeling and optimized Star Schemas to support analytics and BI reporting.

Ingestion & CDC Engineering

- Engineered advanced CDC pipelines using Delta MERGE INTO and Change Data Feed, reducing full refresh workloads by 70%.
- Implemented schema enforcement and controlled schema evolution strategies preventing downstream data contract violations.
- Designed metadata-driven ingestion frameworks reducing dataset onboarding time by 50%.
- Migrated legacy ETL workflows into scalable ELT-based Spark pipelines improving throughput by 40% and reducing compute costs by 25%.

Streaming & Real-Time Processing

- Designed Structured Streaming pipelines using checkpointing, watermarking, and exactly-once semantics ensuring state consistency and fault tolerance.
- Integrated Azure Event Hub ingestion supporting hybrid real-time and micro-batch processing architectures.

Spark Performance & Optimization

- Tuned Spark clusters through executor memory configuration, shuffle partition tuning, and Adaptive Query Execution, improving distributed job efficiency by 32%.
- Applied skew mitigation and Broadcast Join optimization reducing long-tail execution delays by 28%.
- Implemented Delta Z-Ordering, partition optimization, Time Travel, and VACUUM retention policies balancing performance and storage efficiency.

Multi-Cloud Integration

- Integrated AWS ingestion workflows using AWS S3, AWS Glue, and IAM cross-account roles supporting enterprise multi-cloud strategy.

Governance, Security & Data Quality

- Deployed centralized governance using Unity Catalog and RBAC, enforcing fine-grained data access control and lineage visibility.
- Implemented enterprise Data Quality Frameworks including validation rules, schema checks, and anomaly detection improving pipeline reliability by 30%.
- Established audit logging and encrypted storage patterns aligning with enterprise security requirements.

DevOps & Operational Excellence

- Automated CI/CD pipelines using Azure DevOps, reducing release cycles from weekly to daily deployments.
- Established monitoring using Azure Monitor and Log Analytics reducing incident resolution time by 40%.

Analytics & Enablement

- Delivered ML-ready curated datasets, improving model training efficiency by 33%. Partnered with BI

teams to publish optimized Gold-layer datasets to Power BI and Tableau, reducing dashboard refresh times by 30%.

- Designed incremental refresh strategies supporting scalable enterprise BI reporting without cluster degradation.

Leadership

- Led cross-functional architecture reviews aligning Lakehouse modernization roadmap with enterprise transformation initiatives.
- Mentored engineers on distributed computing principles and Spark performance optimization.

Technical Environment: Azure Databricks, Delta Lake, ADLS Gen2, Azure Data Factory, AWS S3, AWS Glue, Unity Catalog, Spark Structured Streaming, Azure DevOps

Devoted Health

Senior Healthcare Data Platform Engineer

AUG 2018 – JAN 2021

Built HIPAA-compliant healthcare data platforms supporting Medicare Advantage analytics, risk adjustment modeling, and regulatory reporting.

Healthcare Data Architecture

- Designed secure ingestion pipelines for claims, member, provider, and EHR datasets processing 5TB+ monthly under strict HIPAA compliance standards.
- Implemented ingestion workflows supporting FHIR and HL7 healthcare data formats standardizing semi-structured clinical data.
- Applied Dimensional Modeling improving healthcare analytical query performance by 38%.

Regulatory, Risk Adjustment & Data Quality

- Engineered risk adjustment pipelines incorporating HCC coding and ICD-10 mappings, improving predictive accuracy by 21%.
- Built curated marts supporting CMS STAR Ratings reporting reducing reporting cycle time by 25%.
- Implemented encryption, tokenization, and PHI masking ensuring HIPAA-aligned distributed processing.
- Developed healthcare-focused Data Quality validation rules ensuring accuracy and completeness of claims and eligibility feeds.

Streaming & Real-Time Processing

- Built real-time pipelines using Kafka and Spark Structured Streaming, reducing eligibility validation latency to minutes.
- Integrated Kafka Schema Registry enforcing schema consistency across streaming systems.

Performance & Reliability

- Reduced ETL failure rates by 45% through automated validation and monitoring frameworks.
- Optimized Spark workloads via partition pruning and broadcast joins reducing processing time by 34%.
- Implemented incremental ingestion strategies improving SLA adherence by 30%. Analytics Enablement
- Delivered curated datasets consumed by Power BI dashboards supporting CMS and population health reporting.

Leadership & Collaboration

- Collaborated with actuarial and compliance teams translating regulatory requirements into scalable

data architecture.

- Provided technical leadership within agile delivery teams ensuring timely healthcare analytics
- deployment.

Technical Environment: Azure Databricks, Spark, Kafka, Delta Lake, ADLS, HIPAA Governance Controls

Cognizant

Data Engineer

AUG 2016 – JUN 2018

Delivered healthcare and fintech data engineering solutions in consulting environments.

- Developed ETL workflows using Informatica, Spark, and Azure Data Factory integrating healthcare transaction systems.
- Designed CDC ingestion using incremental watermarking reducing redundant loads.
- Built Spark batch processing jobs improving performance by 35%.
- Optimized SQL queries reducing reporting runtime by 40%.
- Implemented secure transformation workflows supporting healthcare regulatory standards.
- Designed staging layers standardizing schema mapping across heterogeneous systems.
- Automated ingestion frameworks reducing manual configuration by 60%.
- Tuned Spark workloads via partition pruning improving batch efficiency by 25%.
- Supported fintech, healthcare payment reconciliation pipelines improving transactional accuracy by 29%.
- Assisted cloud migration initiatives transitioning on-prem data warehouses to scalable analytics environments.
- Resolved production data integrity issues within SLA targets.
- Collaborated directly with client stakeholders translating business requirements into technical solutions.

Technical Environment: Informatica, Spark, SQL, Azure Data Factory

EDUCATION

Stevens Institute of Technology

2014 – 2016

Master's degree in Information Technology and Analytics

Stevens Institute of Technology

2010 – 2014

Bachelor's degree in Computer Science